

Decoding Lockbox Solving Stages from Multi-View Mouse and Mechanism Keypoints

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Abstract

Mice solving mechanical lockboxes are usually analysed with two-dimensional keypoints, which cannot say how far a mechanism has travelled or whether the animal was at it when it moved. We place mouse and lockbox in one shared metric three-dimensional frame, recovering camera poses from the rigid lockbox’s known CAD geometry, so that no calibration rig is needed and the scale is fixed in millimetres. A kinematic schema then reads one scalar state per mechanism, and a monotone solving stage, straight out of the geometry. Two of the four extracted onsets match the dataset’s behavioural annotation to within 0.2 s and paw–mechanism distances close to a plausible 3.2 mm, while decoding the same stage from the mouse’s pose alone performs near chance. Re-run on an unseen session without re-training, the extraction still runs, but a mechanism is missed and the onsets drift. We argue that the metric frame, not a learned stage predictor, is the reusable result.

1. Introduction

One way to study problem solving in rodents is to hand the animal a mechanical puzzle. A lockbox presents a sequence of mechanisms, i.e. a lever, a sliding stick, a ball and a sliding door, that must be operated in order before a reward becomes reachable. Its appeal is that the ground truth is mechanical rather than interpretive. At any instant the box is in a well-defined state, and that state changes only when something physically moves it.

Extracting that structure from video is nonetheless usually done in two dimensions. Markerless pose estimators such as DeepLabCut [5] return per-view pixel coordinates, and behaviour is read off them directly. This degrades precisely where the interesting questions live. A pixel distance between a paw and a lever is not a physical distance, since it varies with viewing angle and depth, so “did the animal touch the mechanism” has no well-posed answer. Occlusion is worse still. When a paw vanishes behind the box in

one camera, a 2D analysis cannot recruit the cameras that still see it. Such errors are systematic rather than random, and invisible from within a single view. The seminar topic behind this work asks for a few steps towards a genuinely three-dimensional evaluation of exactly this footage, on the grounds that 3D reasoning can uncover body-part tracking errors a 2D analysis cannot. Given DeepLabCut body-part tracks and DeepLabCut mechanism tracks, the model it proposes is a learned map

$$\begin{aligned}\text{lockbox state} &= f_{\theta}(\text{body-part tracks}) \\ &= f(\text{mechanism positions}),\end{aligned}\tag{1}$$

decoding the state of the box from the mouse’s keypoints on one side and the mechanism’s on the other.

Our central claim is that the second branch of Eq. (1) does not need to be learned at all. The contribution is a pipeline reconstructing the mouse and the mechanisms in a single metric coordinate system without a calibration rig. The lockbox is rigid and its CAD geometry known, so its corners recover the camera poses and fix the scale in millimetres. Within that frame each mechanism’s state follows directly from projecting its tracked keypoint onto its own kinematic axis, giving a displacement in millimetres or an angle in radians that is computed geometrically and checkable against the CAD model. We implement the learned branch as well, and it performs near chance from mouse pose alone, which we report as a negative control rather than a competing method. The information distinguishing a solved from an unsolved mechanism lives in the apparatus, not in the animal’s posture. The mouse reconstruction is instead good for the complementary question of whether a paw was at the mechanism when it moved, which only becomes askable once both reconstructions share a frame. Finally, we test what survives on an unseen session. The solving order transfers, the timing does not.

2. Related Work

The dataset. The Mouse Lockbox Dataset [6] provides roughly 118 hours of grayscale footage of single mice solv-

ing mechanical lockboxes, recorded at 30 Hz from three synchronised perspectives (top-down, side, front) across 12 animals and 11 lockbox variants. It ships per-frame human annotations of mechanism state, which we use as an external reference for everything the pipeline extracts. It does not ship camera calibration. There are no intrinsics, no extrinsics and no metric scale, so the three views are synchronised but not geometrically related to one another.

Pose estimation and 3D lifting. DeepLabCut [5] established the standard recipe of fine-tuning a pretrained backbone on a few hundred hand-labelled frames to track user-defined keypoints, later extended to multiple animals [3]. SuperAnimal [9] removes the labelling step for common taxa by pretraining over pooled datasets, and adds *video adaptation*, refining the model on pseudo-labels harvested from the target video itself. Detection uses a standard Faster R-CNN [7]. All of this is two-dimensional. Anipose [2] supplies the missing half, triangulating per-view detections into 3D trajectories given calibrated cameras. Its *aniposelib* camera group performs the triangulation and reprojection for both reconstructions in this work. It assumes calibration is already solved. No such recording exists here, so we recover pose from known object geometry instead via perspective- n -point [4]. For the learned control we use gradient-boosted trees [1] and a transformer encoder [8].

The gap. The dataset provides synchronised multi-view video and per-frame behavioural annotations, but no calibration, no 3D pose and no machine-readable mechanism state in a metric frame. DeepLabCut and SuperAnimal solve 2D keypoints and *aniposelib* solves triangulation, yet nothing in the existing toolchain puts animal and apparatus into the same metric coordinate system, which is exactly what is needed to ask whether the mouse physically drove a state change rather than merely co-occurring with it. This report closes that gap for one lockbox variant.

3. Method

The pipeline turns three synchronised videos into one metric 3D scene containing both the animal and the apparatus, then reads the solving stage out of it. Figure 1 shows the arrangement. Two detection branches run independently and meet only at the shared calibration, so everything reconstructed afterwards lives in the same coordinate system.

3.1. Multi-view 2D pose

For the mouse we use SuperAnimal detectors off the shelf, *TopViewMouse* for the overhead camera and *Quadruped* for the side and front cameras, each an HRNet-W32 backbone behind a Faster R-CNN detector [7]. Each view is adapted

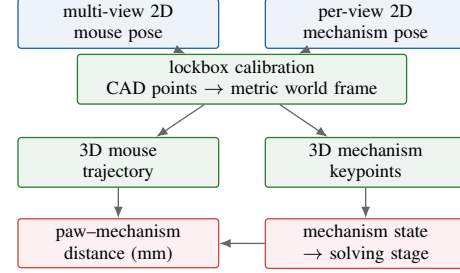


Figure 1. Pipeline overview. Mouse and mechanism keypoints are detected independently but lifted with the same camera group recovered from the rigid lockbox. Sharing the calibration is what makes paw-mechanism distances physically meaningful.

separately on its own video through pseudo-label refinement. The two model families use different keypoint vocabularies, so we map both onto a common 20-point skeleton. Paws exist only in the Quadruped vocabulary and are therefore never seen by the top camera.

Mechanism keypoints cannot come from a pretrained model, since none knows what a lockbox lever is. We define eight mechanism keypoints (one moving point per mechanism plus static references such as the lever pivot) and train one DeepLabCut model per view on hand-labelled frames.

3.2. Calibration from the apparatus

The recordings contain no calibration target, but they do contain a rigid object of exactly known geometry whose CAD model is public. In a frame with the animal away from the box we click six distinctive points (base-plate and mechanism mounting corners) per view and pair them with their CAD coordinates in millimetres, and camera pose follows from perspective- n -point [4]. Intrinsics are not estimated. We assume a pinhole model with no distortion and take the focal length from the camera’s documented sensor pitch and lens. The three cameras form one *aniposelib* camera group shared by every later stage, and because the CAD points are metric, this single step fixes the world frame and its scale.

3.3. Triangulation and gap filling

Merging the per-view detections gives a $(\text{cameras} \times \text{frames} \times \text{keypoints} \times 2)$ tensor, entries below confidence 0.5 discarded, and any keypoint seen by two or more cameras is triangulated directly. This alone is sparse, since the top camera sees no paws and the side and front cameras frequently lose the animal behind the box. A keypoint seen by a single camera is recovered onto that camera’s back-projected ray, which fixes its direction but infers its depth, so we flag such points throughout. Remaining short gaps are then linearly interpolated.

3.4. Mechanism state from a kinematic schema

The step that makes the box readable is a small declarative schema. Each mechanism is described by its moving keypoint, its static references and the one degree of freedom it possesses, in the calibrated world frame. Given the triangulated position $\mathbf{p}_t$ this yields one scalar per frame. For the translating mechanisms the state is the displacement along the rail direction $\hat{\mathbf{a}}$ from the rail origin $\mathbf{o}$,

$$s_t = (\mathbf{p}_t - \mathbf{o}) \cdot \hat{\mathbf{a}}, \quad (2)$$

in millimetres. The ball is unconstrained, so its state is how far it has been carried from its cradle, $s_t = \|\mathbf{p}_t - \mathbf{o}\|$, counting as removed beyond a fixed radius. The lever is measured in the top view alone, since its rotation plane sits nearly edge-on to that camera. Its arm is rigid and hence of constant length, so comparing the apparent distance d_t between the projected pivot and the tracked arm end against the apparent length L of a fully rotated arm gives

$$\theta_t = \arcsin(\text{clip}(d_t/L, 0, 1)), \quad (3)$$

a magnitude on $[0, \pi/2]$, sufficient because the lever is either engaged or not.

A mechanism counts as engaged once its scalar crosses a fixed threshold, debounced over a few consecutive frames and forced into the physical order of the puzzle, since each mechanism unlocks the next. The cumulative index defines a monotone solving stage in $\{0, \dots, 4\}$.

4. Experiments

4.1. Data

The experiments use the LB#2 combined lockbox from the Mouse Lockbox Dataset [6], whose four mechanisms must be solved in a fixed order, namely a lever, a sliding stick, a ball to be removed and a sliding cover. Two sessions of the same animal (mouse 291) on consecutive days are used, both at 30 Hz from the top, side and front cameras. *Scene 1* (18 June 2021, 8190 frames, 273 s) is the session on which the pipeline was built, since the calibration was clicked here and the per-view mechanism detectors trained here. *Scene 2* (17 June 2021, 5670 frames, 189 s) is held out entirely and touched only in Sec. 4.3. Both ship per-frame human state annotations, used throughout as an external reference.

4.2. Experiment 1, building the metric frame

Reconstruction. The three adapted detectors were run over the full session and their 20 mapped keypoints triangulated, followed by the recovery steps of Sec. 3, tracking the provenance of every point so that the three tiers can be told apart. For the mechanisms we labelled 665 frames per view ($\approx 8\%$ of scene 1), selected by combining uniform temporal coverage with dense sampling around moments where

the animal is near a mechanism, and pruned each view’s project to the keypoints actually labelled in that view before training. Camera pose was cross-checked by reprojecting triangulated mouse keypoints, which the calibration was never fitted to.

Interaction analysis. A contact requires a front paw or the nose to come within 30 mm of a mechanism for at least two consecutive frames, searched within ± 6 s of that mechanism’s state change. This predicts nothing. It is a consistency check on the shared frame, and it fails loudly if the two reconstructions do not sit in the same place.

4.3. Experiment 2, transfer to an unseen session

The second experiment asks whether the *geometric* extraction transfers. No predictor is involved, only the schema reading mechanism states out of a reconstruction. Scene 2 is pushed through every stage with no retraining and no re-clicking. The video-adapted weights from scene 1 are applied with adaptation disabled, the scene 1 camera group is reused verbatim, the scene 1 mechanism detectors are run on the new videos, and the same schema and thresholds extract the stage, which is scored against scene 2’s own annotation.

4.4. Explored and discarded

Several avenues were built and measured before being dropped. Table 2 in the appendix summarises them alongside the supporting figures.

5. Results

5.1. Detection, calibration and reconstruction

Video adaptation is the cheapest improvement in the pipeline. In Fig. 2a the mean fraction of top-view keypoints detected above confidence 0.4 rises from 18% to 33%, and post-adaptation traces are continuous over stretches where the original model produced isolated fragments.

The calibration places the three cameras plausibly around the box (Fig. 2b). Reprojecting the six clicked CAD points gives median errors of 5.2, 5.6 and 5.8 px for the side, top and front cameras, a number that flatters the calibration, since those are the points it was fitted to. Reprojecting triangulated mouse keypoints instead, 10 127 observations the calibration never saw, gives a median of 11.9 px. Direct triangulation of the mouse is indeed sparse, with only 14.4% of keypoint-frames seen by two or more cameras, since the top camera does not observe paws and the side and front cameras lose the animal behind the box. Single-view recovery raises this to 49.7% and short-gap interpolation to 58.6%, so roughly a third of the reconstruction has an inferred rather than a measured depth. The resulting poses

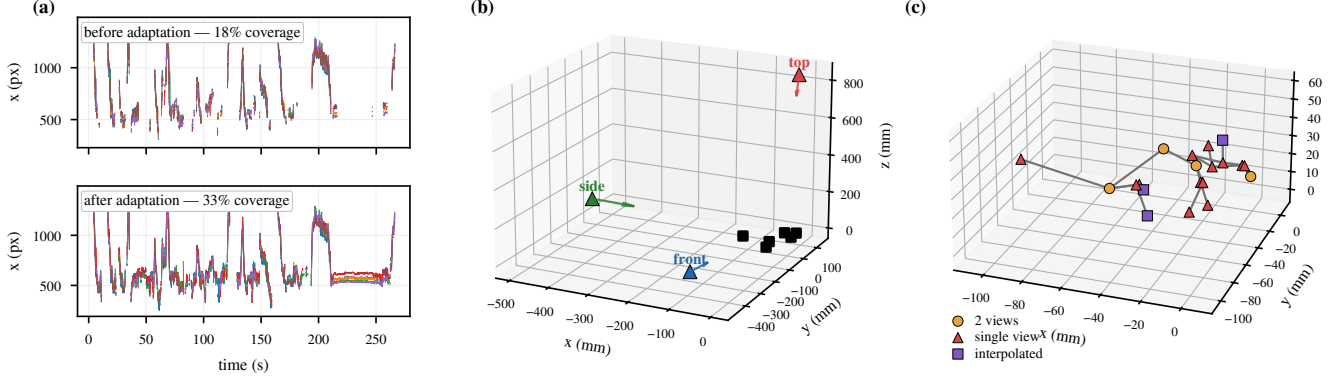


Figure 2. Reconstruction quality on scene 1. **(a)** Horizontal image coordinate of six head keypoints in the top view before and after video adaptation. Missing segments are frames that fell below the confidence cutoff. **(b)** Recovered camera poses and the six CAD calibration points (black squares) in the metric world frame, with arrows marking the optical axes. **(c)** Triangulated mouse pose at frame 171, all 20 keypoints present, coloured by provenance.

Table 1. Extracted stage onsets on scene 1 against the dataset’s human annotation.

Mechanism	Annotated	Extracted	Δ
lever	179.5 s	179.8 s	+0.2 s
stick	192.8 s	186.7 s	−6.1 s
ball	193.2 s	208.8 s	+15.6 s
sliding door	211.5 s	211.4 s	−0.1 s

are nonetheless anatomically coherent (Fig. 2c). The mechanisms are far denser, the lockbox being rigid and always in frame, at 78.0% from two-view triangulation alone and 82.8% after recovery.

5.2. Mechanism state extraction

Figure 3 shows the four extracted states and the resulting stage. The mechanisms behave as the schema predicts. The stick advances in discrete steps before jumping 30 mm at 187 s, the ball holds its cradle radius until carried away past 100 mm, and the lever rests near 0.23 rad before flipping to $\pi/2$.

Against the human annotation (Tab. 1) the lever and sliding door are recovered essentially exactly. The stick fires 6.1 s early and the ball 15.6 s late, and neither is a reconstruction failure. The stick crosses its 5 mm threshold while still being handled, before the annotator judged it solved. The ball is hidden by the animal while being carried, so it only registers as removed once the mouse lets go and the keypoint is detected again.

5.3. Predicting the lockbox state from keypoints

The seminar’s proposed model, Eq. (1), asks for the lockbox state as a learned function of body-part tracks on one side and of mechanism positions on the other. We trained both

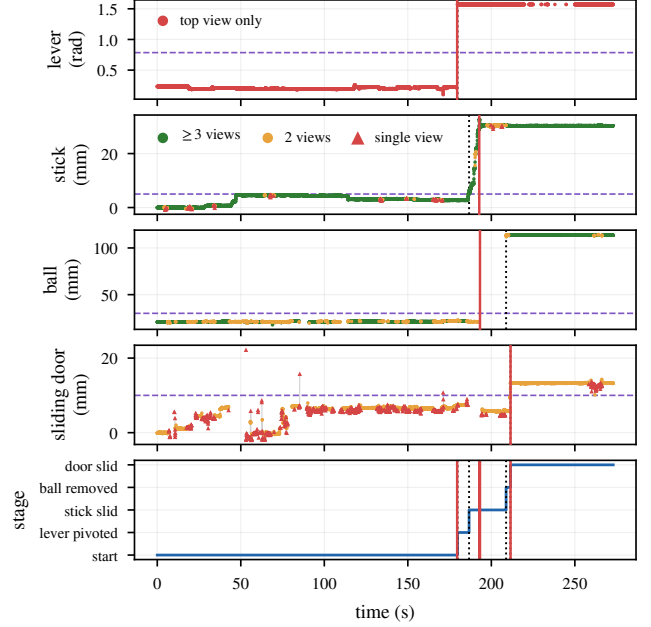


Figure 3. Extracted mechanism states on scene 1. Marker colour encodes how many cameras contributed to each point, and the lever is extracted from the top view alone. Dashed lines are engagement thresholds, dotted vertical lines the detected onsets, solid red lines the human annotation. The bottom panel shows the resulting monotone solving stage.

branches against the same reference, four feature sets (lockbox and mouse keypoints, each in 2D across the three views and in 3D after triangulation) crossed with three model families, evaluated with blocked cross-validation and a held-out final strip of the session.

Figure 4a answers it, and the two branches are not comparable. From lockbox keypoints the stage is recovered al-

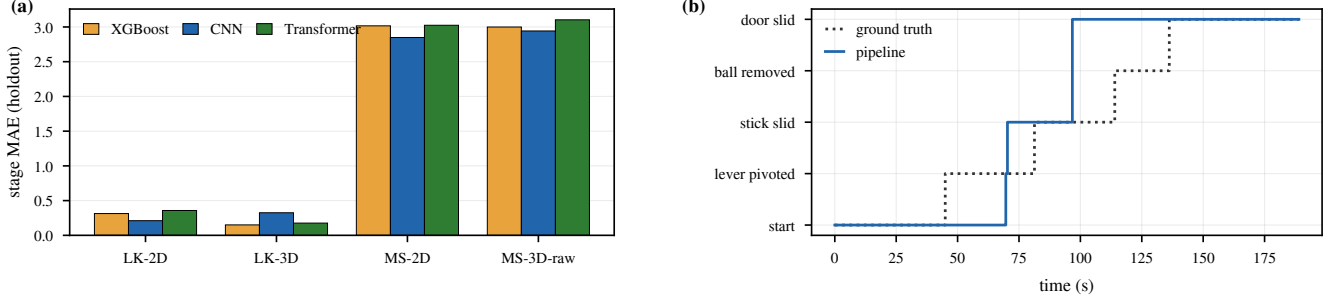


Figure 4. **(a)** Holdout stage MAE per feature set and model family (lower is better). Lockbox keypoints decode the stage, mouse keypoints do not, regardless of model. **(b)** Unseen session, with no retraining or re-calibration. The order of the stages is recovered, but the transitions are compressed into a much shorter interval than the annotation.

most perfectly. 3D lockbox keypoints with gradient boosted trees reach a stage MAE of 0.150 and 93.2% per-frame accuracy, with 2D lockbox features close behind at 0.211 MAE. From the mouse it is not. The best mouse configuration reaches 2.85 MAE and 16.4% accuracy, close to chance for a five-class monotone target. The gap of more than an order of magnitude holds across all three model families, so it is a property of the features rather than of any one model, and temporal context does not rescue it, since the windowed CNN and transformer land within a few percent of the per-frame baseline. Two caveats bound the left-hand number. On scene 1 the holdout strip is the final 10% of the same video and the labels were themselves produced by the extraction pipeline, so 93.2% is a within-session consistency figure rather than a generalisation figure. The mechanism branch also does not need learning in the first place, since the same quantity follows directly from the geometry of Eqs. (2) and (3).

5.4. Paw–mechanism interaction

Applying a 30 mm contact criterion within ± 6 s of each state change, contact precedes the change for two of four mechanisms, the lever by 1.23 s (front left paw) and the sliding door by 0.67 s (nose). The ball registers 0.13 s after its change, within the uncertainty of the onset itself, and the stick registers no contact at all, its closest in-window approach being 75.5 mm. Neither miss indicates a broken frame. The stick’s is a keypoint placement choice, since the mechanism is represented by its knob whereas the animal manipulates the shaft further along, so the paw is genuinely at the mechanism but far from the point standing for it. The ball’s is missing data, with only 0–1.1% of frames in its pre-change window having both points reconstructed at once.

5.5. Transfer to an unseen session

Re-running the pipeline on scene 2 without retraining reproduces the structure of the solve but not its timing (Fig. 4b). Per-frame stage accuracy is 60.2% with a stage MAE of 0.489, and the pipeline reaches the final stage as the anno-

tation does, but the onsets drift badly, the lever by 24.7 s and the sliding door by 39.5 s, and the ball is never detected at all. The cause is the mechanism detection itself. Lockbox keypoints transfer poorly to a session they were not trained on, so 3D coverage falls from 82.8% to 65.1% and the extracted states become patchy enough that the thresholds fire at the wrong moments. Because the stage machine enforces the puzzle’s order, one early trigger then carries into every onset after it.

6. Discussion

Learning the state is the wrong instrument. A single session cannot support a learned stage predictor, and it does not need to. Training and holdout come from the same video, and the labels were produced by the very pipeline the decoder is scored against, so a high within-session score mostly confirms that the features are self-consistent. Nor can one session separate what a model learned about lockboxes from what it learned about this recording, and a temporal window over a video containing exactly one solving sequence adds context, not examples. The deeper point is that the comparison in Fig. 4a was never symmetric. The mechanism branch of Eq. (1) is solved geometrically, needs no training at all, and recovers two of four onsets to within 0.2 s, while the body-part branch performs near chance under every architecture. That is unsurprising in hindsight, since a mouse at a solved lever and a mouse at an unsolved lever look nearly identical, and the distinguishing information lives in the apparatus rather than the posture. Reading the state off the mechanism is also interpretable in a way a regression output is not, since a displacement in millimetres along a known axis can be checked against the CAD model. The productive role for the mouse reconstruction is therefore not to predict the state but to explain it, which is what the paw–mechanism analysis does.

Accurate, but not scalable. Everything the pipeline does well rests on manual work, the calibration clicks, the per-

view labelling, and the schema’s axes and thresholds, all of which must be redone for every lockbox variant and whenever the rig moves. Against 11 variants and 12 animals this is prohibitive.

7. Conclusion

The reusable outcome of this work is not a model but a frame. The lockbox is rigid and its CAD geometry known, so the apparatus itself replaces the checkerboard the recordings never contained, and calibrating against it reconstructs the mouse and the four mechanisms in one metric coordinate system, turning paw–mechanism separation into millimetres and mechanism travel into physical displacement. Within that frame the solving stage is read off geometrically, matching human annotation to within 0.2 s for two of four mechanisms, with distances closing to a physically plausible 3.2 mm. The decoding question resolves asymmetrically. Lockbox keypoints determine the stage (93.2% within-session) while mouse keypoints essentially do not (16.4% at best) across three model families and both 2D and 3D features. Since the mechanism state is directly computable from geometry, the learned branch is not merely weaker but redundant. Transferring the pipeline to an unseen session without retraining preserves the ordering of the solve (60.2%) while losing its timing, marking the honest limit of what one calibrated, hand-labelled session supports. The bottleneck throughout is manual effort, not model capacity.

8. Future Work

The clearest next step follows from the finding that geometry outperforms learning here. The lockbox geometry should be used *directly* rather than through learned keypoints. The CAD model of every mechanism is known exactly and the cameras are calibrated, so a mechanism’s configuration could be estimated by rendering its mesh at a hypothesised state and comparing against the observed frames, optimising the single degree of freedom directly. That removes the hand-labelled keypoints entirely, since the pose is fitted rather than regressed, and yields a state in physical units by construction, with a render-versus-image residual as a built-in confidence measure. It also removes the arbitrary choice of where on a mechanism to place a keypoint, which is exactly what caused the stick’s contact to be missed in Sec. 5. With the labelling bottleneck gone, extraction could be scaled to all lockbox variants and animals in the dataset, which is the only way to establish whether solving behaviour generalises across mechanisms and individuals rather than within a single session.

References

- [1] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pages 785–794, 2016. 2

Table 2. Approaches evaluated and cut. None is carried by the reported pipeline.

Approach	Outcome
IR pre-processing (CLAHE, 9 variants)	no measurable detection gain over raw frames (Fig. 5)
Top-view variant search	best variant matched the raw top view
Mouse-based calibration	a deformable subject gives unstable pose and no metric scale
NN pose inpainting	full coverage, but matches interpolation with no downstream gain (Figs. 6 and 7)
Pooled model	lockbox confuses keypoints a given view never observes

- [2] Pierre Karashchuk, Katie L. Rupp, Evyn S. Dickinson, Sarah Walling-Bell, Elischa Sanders, Eiman Azim, Bingni W. Brunton, and John C. Tuthill. Anipose: A toolkit for robust markerless 3D pose estimation. *Cell Reports*, 36(13):109730, 2021. 2
- [3] Jessy Lauer, Mu Zhou, Shaokai Ye, William Menegas, Steffen Schneider, Tanmay Nath, Mohammed Mostafizur Rahman, Valentina Di Santo, Daniel Soberanes, Guoping Feng, Venkatesh N. Murthy, George Lauder, Catherine Dulac, Mackenzie W. Mathis, and Alexander Mathis. Multi-animal pose estimation, identification and tracking with DeepLabCut. *Nature Methods*, 19:496–504, 2022. 2
- [4] Vincent Lepetit, Francesc Moreno-Noguer, and Pascal Fua. EPnP: An accurate O(n) solution to the PnP problem. *IJCV*, 81(2):155–166, 2009. 2
- [5] Alexander Mathis, Pranav Mamidanna, Kevin M. Cury, Taiga Abe, Venkatesh N. Murthy, Mackenzie W. Mathis, and Matthias Bethge. DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. *Nature Neuroscience*, 21(9):1281–1289, 2018. 1, 2
- [6] Patrik Reiske, Marcus Boon, Niek Andresen, et al. Mouse lockbox dataset: Behavior recognition for mice solving lockboxes, 2025. 1, 3
- [7] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *NeurIPS*, 2015. 2
- [8] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, 2017. 2
- [9] Shaokai Ye, Anastasiia Filippova, Jessy Lauer, Steffen Schneider, Maxime Vidal, Tian Qiu, Alexander Mathis, and Mackenzie W. Mathis. SuperAnimal pretrained pose estimation models for behavioral analysis. *Nature Communications*, 15:5165, 2024. 2

A. Discarded approaches

Supporting evidence for the approaches summarised in Tab. 2. None of these is used by the reported pipeline.

IR pre-processing. Nine contrast treatments of the infrared top-down frames were compared (Fig. 5). CLAHE

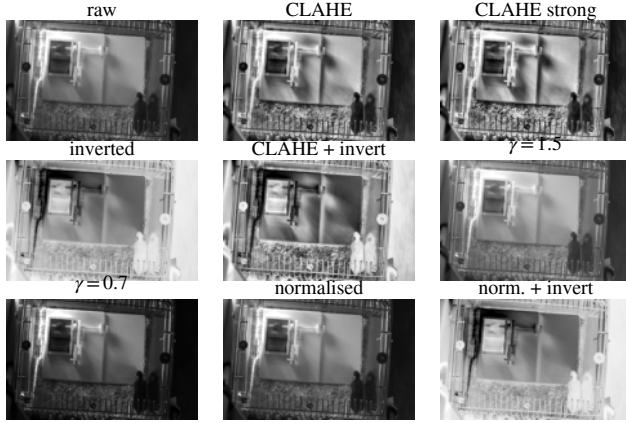


Figure 5. Contrast treatments evaluated on the infrared top-down view. Visible contrast improves, detection coverage does not.

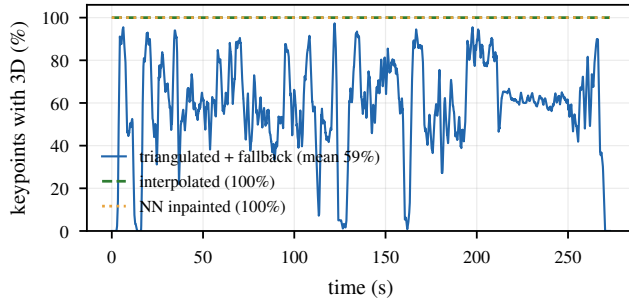


Figure 6. 3D coverage of the mouse pose over the session. Triangulation plus fallback leaves substantial gaps, while both interpolation and the neural inpainter close them completely and to equivalent effect.

visibly improves apparent contrast, but detection coverage after video adaptation was indistinguishable from the raw frames, so the extra encoding pass was dropped.

Neural pose inpainting. A small windowed transformer was trained to reconstruct masked keypoints from the visible ones, filling every gap in the 3D mouse pose (Fig. 6). It reaches full coverage, but so does linear interpolation, and the two agree closely on the filled positions. Adding the inpainted features to the decoders changes nothing. Figure 7 extends Fig. 4a with the inpainted mouse features (MS-3D-nn) and their combination with the lockbox features (LK + MS-3D-nn). The inpainted mouse features remain close to chance, and the combination does not exceed the lockbox features alone, so the mouse contributes nothing the lockbox does not already provide.

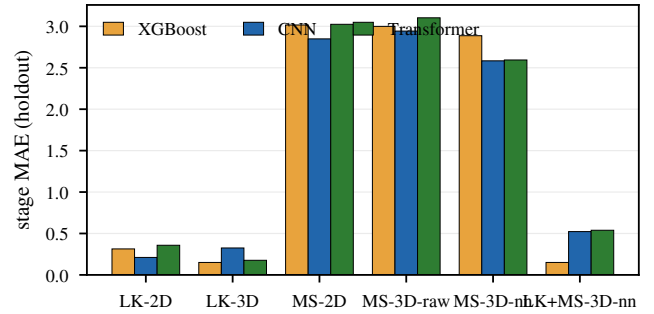


Figure 7. Holdout stage MAE including the inpainted mouse features and the combined feature set. Neither improves on the lockbox features alone.